Chapter 21

Differential Abundance Analysis Using Linear Models and Empirical Bayes Moderation

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Abstract

Differential abundance analysis constitutes the central inferential milestone in quantitative mass spectrometry (MS), bridging upstream peptide identification, feature extraction, and normalization with downstream biological pathway interpretation. In bottom-up proteomics and untargeted metabolomics, thousands of biomolecular entities ($p \approx 10^3$–10^4) are profiled across a limited cohort of biological replicates ($n \approx 3$–10 per phenotype). Under such severe small-sample high-dimensional regimes ($p \gg n$), conventional statistical tests (e.g., standard Student's t-tests and ordinary ANOVA) suffer catastrophic loss of statistical power and variance instability: features with underestimated sample variances yield spuriously inflated test statistics, manufacturing hundreds of false-positive discoveries out of measurement noise.

This chapter develops the theoretical, mathematical, and practical foundations of empirical Bayes variance moderation pioneered by Gordon Smyth (`limma`) and adapted for quantitative mass spectrometry. We formulate the high-dimensional linear regression framework, explicate the inverse-gamma prior distribution for feature-specific error variances, derive the closed-form moderated t-statistic and log-odds (B-statistic), and establish rigorous false discovery rate (FDR) control via the Benjamini–Hochberg and Storey procedures. We provide complete algebraic formulations for two-group, factorial, paired, and multi-site blocked designs. The methodology is operationalized through a hands-on, step-by-step computational R laboratory utilizing the benchmark 11-site Data-Independent Acquisition (DIA) multi-laboratory reproducibility study (PXD004886) containing synthetic AQUA spike-in positive controls. The chapter concludes with an audit of the ten deadly sins of differential omics, comprehensive multi-tier exercises, and fully worked-out step-by-step mathematical and algorithmic solutions.

Keywords: Differential Abundance, Empirical Bayes, Linear Models, `limma`, False Discovery Rate, Mass Spectrometry, Proteomics, Volcano Plot, Contrast Matrix.

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21.1 The Importance of Differential Abundance in Mass Spectrometry

High-throughput mass spectrometry measures the relative or absolute abundance of thousands of proteins, peptides, or metabolites across biological states [12, 13, 26]. Whether interrogating a cancer biopsy cohort, a drug-treatment time course, or microbial adaptations to nutrient deprivation, the primary analytical objective is almost universally comparative: *Which molecular entities undergo statistically genuine, biologically meaningful changes in abundance between phenotypic conditions?*

In classical univariate statistics, testing the equality of two population means $\mu_1 = \mu_2$ relies on Student's two-sample t-test. In modern omics, however, the analyst faces four physical and statistical realities inherent to mass spectrometry:

1. **The High-Dimensional Small-Sample Regime ($p \gg n$):** Modern LC-MS/MS platforms routinely quantify $5,000$ to $10,000$ protein groups or $1,000$ to $3,000$ metabolite features across only 3 to 5 biological replicates per group [11]. With $d_g = 4$ residual degrees of freedom, sample variance estimates fluctuate erratically across several orders of magnitude purely by sampling error [1, 2].

2. **Dynamic Range and Mean-Variance Dependence:** MS ion intensities span five to six orders of magnitude (10^3 to 10^9). Lower-abundance features exhibit higher technical coefficients of variation (CV) and elevated rates of missingness (Missing Not At Random, MNAR), whereas high-abundance housekeeping proteins display lower variance [15, 19].

3. **Multi-Layered Batch and Nuisance Structure:** Instrument drift, column maintenance, electrospray emitter fouling, and multi-day digestions introduce technical variance that frequently eclipses subtle biological signals unless explicit blocking factors or pairing constraints are incorporated into the design matrix [23, 24].

4. **Multiplicity and the Multiple Comparison Problem:** Testing $5,000$ proteins simultaneously at an unadjusted $\alpha = 0.05$ expects 250 false-positive discoveries by chance alone under the complete null hypothesis, necessitating rigorous multiple testing control [5, 7].

To isolate biological variation from technical noise, the analytical pipeline must integrate linear modeling with information borrowing across the entire proteome or metabolome [27, 28]. Empirical Bayes variance shrinkage provides an optimal trade-off: it shrinks volatile individual variance estimates toward a global consensus prior, restoring statistical power while maintaining rigorous false discovery rate control.

21.2 Mathematical Formulations and Statistical Foundations

21.2.1 The High-Dimensional Linear Model

Let y_{gi} denote the normalized, $\log_2$-transformed abundance of feature g ($g = 1, \ldots, G$) in sample i ($i = 1, \ldots, N$). For each feature g, the relationship between abundance and experimental design is modeled as a linear regression:

$$\mathbf{y}_g = \mathbf{X}\beta_g + \varepsilon_g, \quad \varepsilon_g \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{W}_g^{-1}) \tag{21.1}$$

where:

- $\mathbf{y}_g = (y_{g1}, y_{g2}, \ldots, y_{gN})^T$ is the $N \times 1$ vector of observed $\log_2$ abundances for feature g.
- $\mathbf{X}$ is the known $N \times P$ design matrix specifying sample covariates, phenotypic groups, and technical blocking factors ($P < N$).

- $\beta_g = (\beta_{g1}, \beta_{g2}, \ldots, \beta_{gP})^T$ is the $P \times 1$ vector of unknown regression coefficients representing baseline abundances, condition effects, or interaction terms.
- $\mathbf{W}_g$ is an $N \times N$ known positive-definite weight matrix (typically the identity matrix $\mathbf{I}_N$, or precision weights incorporating intensity-dependent trend structures as in `voom` [4]).
- σ_g^2 is the unknown residual error variance for feature g.

The weighted least squares (WLS) estimator of β_g is:

$$\hat{\beta}_g = (\mathbf{X}^T \mathbf{W}_g \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W}_g \mathbf{y}_g \tag{21.2}$$

Conditional on σ_g^2, the coefficient estimator follows a multivariate normal distribution:

$$\hat{\beta}_g \mid \sigma_g^2 \sim \mathcal{N}\left(\beta_g, \sigma_g^2 (\mathbf{X}^T \mathbf{W}_g \mathbf{X})^{-1}\right) \tag{21.3}$$

21.2.2 Contrasts of Biological Interest

A biological hypothesis rarely corresponds to whether an individual model parameter $\beta_{gp} = 0$. Instead, researchers test linear combinations of parameters known as *contrasts*. Let $\mathbf{c}_j$ be a $P \times 1$ contrast vector. The contrast parameter γ_{gj} is defined as:

$$\gamma_{gj} = \mathbf{c}_j^T \beta_g \tag{21.4}$$

The unbiased point estimator of the contrast is:

$$\hat{\gamma}_{gj} = \mathbf{c}_j^T \hat{\beta}_g \tag{21.5}$$

The sampling variance of $\hat{\gamma}_{gj}$ is given by:

$$\mathrm{Var}(\hat{\gamma}_{gj} \mid \sigma_g^2) = \sigma_g^2 v_{gj}, \quad \text{where } v_{gj} = \mathbf{c}_j^T (\mathbf{X}^T \mathbf{W}_g \mathbf{X})^{-1} \mathbf{c}_j \tag{21.6}$$

The scalar v_{gj} is the unscaled variance coefficient, fully determined by the experimental design and contrast specification.

21.2.3 The Failure of Ordinary t-Statistics at Small Sample Size

The classical unbiased estimator of the residual variance σ_g^2 is the sample residual variance s_g^2:

$$s_g^2 = \frac{1}{d_g} (\mathbf{y}_g - \mathbf{X}\hat{\beta}_g)^T \mathbf{W}_g (\mathbf{y}_g - \mathbf{X}\hat{\beta}_g) \tag{21.7}$$

where $d_g = N - \mathrm{rank}(\mathbf{X})$ represents the residual degrees of freedom. The standard Student's t-statistic tests the null hypothesis $H_0 : \gamma_{gj} = 0$:

$$t_{gj} = \frac{\hat{\gamma}_{gj}}{s_g \sqrt{v_{gj}}} \tag{21.8}$$

Under the null hypothesis, $t_{gj} \sim t_{d_g}$. However, in MS experiments with $N_1 = N_2 = 3$ replicates, $d_g = 6 - 2 = 4$. Under four degrees of freedom:

- The distribution of s_g^2 has a heavy right tail and high density near zero: $\mathrm{Var}(s_g^2) = \frac{2\sigma_g^4}{d_g} = \frac{\sigma_g^4}{2}$.
- Features whose sample variance happens to be close to zero by chance ($s_g^2 \approx 0$) yield massive t-statistics even when the effect size $\hat{\gamma}_{gj}$ is trivial (false positive inflation).
- Conversely, genuine biological changes with slightly overestimated sample variance are penalized by excessively small test statistics (false negative inflation).

21.2.4 Smyth's Empirical Bayes Hierarchical Prior

To stabilize variance estimation, Gordon Smyth (2004) introduced a hierarchical empirical Bayes model that links variance estimates across all G features [1]. The true unknown feature variance σ_g^2 is assumed to follow a scaled inverse chi-square prior distribution:

$$\frac{1}{\sigma_g^2} \sim \frac{1}{d_0 s_0^2}\chi^2_{d_0} \iff \sigma_g^2 \sim \text{Scale-Inv-}\chi^2(d_0, s_0^2) \tag{21.9}$$

where:

- s_0^2 is the global prior variance, representing the expected typical variance across all quantified features.
- d_0 is the prior degrees of freedom, quantifying the degree of belief in the prior distribution.

The sample residual variance s_g^2 conditional on the true variance σ_g^2 follows:

$$s_g^2 \mid \sigma_g^2 \sim \frac{\sigma_g^2}{d_g}\chi^2_{d_g} \tag{21.10}$$

By Bayes' theorem, combining the likelihood (21.10) and the conjugate prior (21.9) yields the posterior distribution of σ_g^2:

$$\sigma_g^2 \mid s_g^2 \sim \text{Scale-Inv-}\chi^2(d_0 + d_g, \tilde{s}_g^2) \tag{21.11}$$

where the **moderated residual variance** $\tilde{s}_g^2$ is the posterior mean:

$$\tilde{s}_g^2 = \frac{d_0 s_0^2 + d_g s_g^2}{d_0 + d_g} \tag{21.12}$$

Takeaway Summary

The Shrinkage Mechanism: The moderated variance $\tilde{s}_g^2$ is a precision-weighted average of the empirical feature-specific variance s_g^2 and the global consensus variance s_0^2:

- If d_g is very small (e.g., $d_g = 4$) and $d_0 = 12$, the prior dominates: $\tilde{s}_g^2 \approx s_0^2$. Features cannot artificially achieve significance through anomalously small variances.
- As sample size increases ($d_g \to \infty$), the empirical data dominates: $\tilde{s}_g^2 \to s_g^2$, reverting smoothly to standard ordinary least squares.
- When outlier variances occur due to hypervariable proteins, Phipson et al. (2016) developed robust hyperparameter estimation (`robust = TRUE`), protecting the empirical Bayes prior against distortion by heavy-tailed distributions [3].

21.2.5 The Moderated t-Statistic and the B-Statistic

Replacing the noisy sample variance s_g in Eq. (21.8) with the moderated standard deviation $\tilde{s}_g$ defines the **moderated t-statistic**:

$$\tilde{t}_{gj} = \frac{\hat{\gamma}_{gj}}{\tilde{s}_g \sqrt{v_{gj}}} \tag{21.13}$$

Theorem (Smyth, 2004): Under the null hypothesis $H_0 : \gamma_{gj} = 0$, the marginal distribution of the moderated t-statistic is an exact Student's t-distribution with augmented degrees of freedom:

$$\tilde{t}_{gj} \sim t_{d_0+d_g} \tag{21.14}$$

The degrees of freedom are augmented from d_g to $d_0 + d_g$. In an experiment with $d_g = 4$ and prior degrees of freedom $d_0 = 20$, the test behaves as if it had 24 degrees of freedom, dramatically sharpening the critical value cutoff and statistical power.

The B-Statistic: Posterior Log-Odds

In addition to the p-value derived from Eq. (21.14), `limma` computes the B-statistic, representing the posterior log-odds of differential expression versus the null hypothesis:

$$B_{gj} = \ln\left(\frac{P(\gamma_{gj} \neq 0 \mid \hat{\gamma}_{gj}, s_g^2)}{P(\gamma_{gj} = 0 \mid \hat{\gamma}_{gj}, s_g^2)}\right) \tag{21.15}$$

When $B_{gj} = 0$, the feature is equally likely to be differentially abundant as non-differentially abundant ($50/50$ odds). A value $B_{gj} > 3$ indicates strong posterior evidence for true differential abundance ($e^3 \approx 20:1$ odds).

21.2.6 Multiple Testing and the False Discovery Rate

Testing $G = 5,000$ proteins simultaneously at significance level $\alpha = 0.05$ without adjustment expects $5,000 \times 0.05 = 250$ false discoveries by chance under the complete null hypothesis. In discovery-oriented mass spectrometry, controlling the **False Discovery Rate (FDR)** is standard practice.

Let $H_{(1)}, H_{(2)}, \ldots, H_{(G)}$ denote the null hypotheses corresponding to sorted raw two-tailed p-values:

$$p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(G)} \tag{21.16}$$

The **Benjamini–Hochberg (1995) step-up procedure** controls the FDR at nominal level $q^* \in (0,1)$ by finding the largest index k such that:

$$k = \max\left\{ i \in \{1,\ldots,G\} : p_{(i)} \leq \frac{i}{G} q^* \right\} \tag{21.17}$$

and rejecting all hypotheses $H_{(1)}, \ldots, H_{(k)}$ [5]. The adjusted p-value (FDR q-value) for the i-th ordered feature is computed monotonically as:

$$q_{(i)} = \min_{j \geq i}\left\{ \min\left(1, \frac{G}{j} p_{(j)}\right) \right\} \tag{21.18}$$

21.3 Design Matrices and Contrast Architectures

The design matrix $\mathbf{X}$ maps observed intensities to experimental parameters. Specifying the design matrix and contrasts correctly is essential to avoid statistical errors.

21.3.1 Parameterization Paradigms: Means vs. Reference Cell

Consider an experiment comparing two conditions: Control (C) and Treatment (T), each with $n = 3$ replicates.

Table 21.1: Comparison of Multiple Testing Methodologies in Mass Spectrometry

Metric / Method	Formal Criterion Controlled	Statistical Power	Typical Omics Application
Bonferroni	Family-Wise Error Rate: $P(V \geq 1) \leq \alpha$	Extremely low	Confirmatory clinical trials; single-target diagnostics
Benjamini–Hochberg [5]	Expected FDR: $\mathbb{E}[V/\max(R,1)] \leq q^*$	High	Standard discovery proteomics and metabolomics
Benjamini–Yekutieli [6]	FDR under arbitrary dependence structures	Conservative	Highly correlated lipidomics/co-expression modules
Storey's q-value [7]	Positive FDR: $\mathbb{E}[V/R \mid R > 0]$	Highest (estimates π_0)	Large-scale cohorts with well-calibrated p-values
Local FDR [8]	Posterior probability $P(H_0 \mid z_g)$	Nuanced	Decoupling effect size from test significance

1. Group-Means Parameterization (`~ 0 + Group`)

The model contains no intercept; each column corresponds to the mean $\log_2$ abundance of a group:

$$\mathbf{X}_{\text{means}} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mu_{\text{Control}} \\ \mu_{\text{Treatment}} \end{pmatrix} \tag{21.19}$$

To test the difference $\mu_{\text{Treatment}} - \mu_{\text{Control}}$, the contrast vector is explicitly:

$$\mathbf{c} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \implies \gamma = \mathbf{c}^T \beta = \mu_{\text{Treatment}} - \mu_{\text{Control}} \tag{21.20}$$

2. Reference-Cell Parameterization (`~ Group`)

The model includes an intercept representing baseline control abundance, and a slope representing the treatment effect:

$$\mathbf{X}_{\text{ref}} = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} \mu_{\text{Control}} \\ \beta_{\text{Treatment}} \end{pmatrix} = \begin{pmatrix} \mu_{\text{Control}} \\ \mu_{\text{Treatment}} - \mu_{\text{Control}} \end{pmatrix} \tag{21.21}$$

The treatment effect is estimated directly by the second coefficient $\beta_{\text{Treatment}}$, tested with $\mathbf{c} = (0,1)^T$.

21.3.2 Factorial Interaction Designs (2×2)

Consider a factorial design with Genotype (WT, KO) and Drug Treatment (Veh, Drug). The group-means approach defines four condition groups: The interaction contrast tests whether the drug effect in Knockout differs from the drug effect in Wild-Type (synergy vs antagonism):

$$\mathbf{c}_{\text{interaction}} = (\mu_{\text{KO.Drug}} - \mu_{\text{KO.Veh}}) - (\mu_{\text{WT.Drug}} - \mu_{\text{WT.Veh}}) \tag{21.22}$$

21.3.3 Paired and Blocked Multi-Site Designs

When samples are matched by subject, batch, or multi-laboratory site, the experimental design must isolate between-block variation [23, 24]. Consider the 11-site DIA reproducibility benchmark (PXD004886) [10], where each laboratory measured Group A and Group B:

$$y_{si} = \mu + \beta_{\text{Group}} \cdot \text{Group}_{si} + \alpha_{\text{Site}(s)} + \varepsilon_{si} \tag{21.23}$$

By assigning a separate dummy parameter to each site in '`~ 0 + group + site`', the model tests `groupB - groupA` strictly *within each site*, averaging the within-laboratory log-fold changes and eliminating inter-site technical baseline shifts.

21.4 Hands-On R Lab: Step-by-Step Differential Testing

This laboratory guides the reader through a complete, reproducible differential abundance workflow on the 11-site DIA reproducibility dataset (PXD004886) [10]. The dataset contains 4,517 quantified protein groups, including synthetic AQUA30 heavy-peptide spike-in standards spiked at known concentrations into Group B.

21.4.1 Lab Step 1: Environment Setup and Data Ingestion

We load the required analytical libraries and import the annotated differential abundance dataset:

R Lab Listing 1: Environment Setup and Dataset Ingestion [R 4.6 / Bioconductor]

```r
library(limma)
library(dplyr)
library(ggplot2)
library(ggrepel)

# Ingest the 11-site DIA benchmark results
de_path <- "data/pxd004886/DE_results_annotated.csv"
if (!file.exists(de_path)) {
  de_path <- "../data/pxd004886/DE_results_annotated.csv"
}
de_results <- read.csv(de_path, stringsAsFactors = FALSE)

cat(sprintf("Successfully ingested %d protein groups across 11 sites.\n",
            nrow(de_results)))
```

```
Successfully ingested 4517 protein groups across 11 sites.  First 3 features:  Protein Genename log2FC P.Value
adj.P.Val 1 P04217 A1BG -0.052841 0.78451290 0.89230145 2 P01009 A1AT 0.081240 0.61204921 0.77891230 3 P02647 APOA1
0.124890 0.45102914 0.64321094
```

21.4.2 Lab Step 2: Diagnostic Inspection and Quality Assessment

Prior to interpreting differential candidates, we evaluate the distribution of test statistics and p-values. A well-calibrated statistical model produces a flat uniform distribution under the null hypothesis, with an anti-conservative spike near zero reflecting true differential targets:

R Lab Listing 2: Diagnostic P-Value Distribution Profiling [R 4.6 / Bioconductor]

```r
p_hist <- ggplot(de_results, aes(x = P.Value)) +
  geom_histogram(breaks = seq(0, 1, by = 0.05), fill = "#005A9C",
                 color = "white", alpha = 0.85) +
  geom_hline(yintercept = nrow(de_results) / 20, linetype = "dashed",
             color = "#B91C1C", linewidth = 0.7) +
  labs(title = "P-Value Distribution Diagnostic",
       subtitle = "Flat null expectation with enrichment of true positives near p < 0.01",
       x = "Raw P-Value", y = "Frequency") +
  theme_minimal(base_size = 11)
```

Methodological Note

Diagnostic Interpretation: When raw p-values exhibit a 'U-shaped' distribution or an unexpected peak near 1.0, model assumptions have failed. In mass spectrometry, this typically signals unmodeled batch effects, severe intensity-dependent missingness (MNAR), or correlation between replicates not captured by the design matrix [15, 19].

21.4.3 Lab Step 3: Specifying the Model Matrix and Contrasts

In an end-to-end workflow starting from normalized log-expression matrix `expr_mat` ($G \times N$), we construct the paired multi-site design matrix:

R Lab Listing 3: Specifying Paired Multi-Site Model Matrix [R 4.6 / Bioconductor]

```r
n_sites <- 11
sample_metadata <- data.frame(
  sample_id = paste0("S", rep(1:n_sites, each = 2), "_", rep(c("A", "B"), n_sites)),
  site      = factor(rep(paste0("Site_", sprintf("%02d", 1:n_sites)), each = 2)),
  group     = factor(rep(c("A", "B"), n_sites))
)

# Build design matrix with group means and site blocking
design <- model.matrix(~ 0 + group + site, data = sample_metadata)
colnames(design) <- make.names(colnames(design))

# Construct contrast vector for Group B vs Group A
contrast_matrix <- makeContrasts(
  GroupB_vs_GroupA = groupB - groupA,
  levels = design
)
print(contrast_matrix)
```

21.4.4 Lab Step 4: Model Fitting and Empirical Bayes Moderation

We fit linear models feature-by-feature using weighted least squares via `lmFit()`, compute contrasts with `contrasts.fit()`, and shrink residual variances toward the inverse-gamma prior via `eBayes()`:

R Lab Listing 4: Model Fitting with Weighted Least Squares and eBayes [R 4.6 / Bioconductor]

```r
# Fit linear models across all quantified proteins
fit <- lmFit(expr_mat, design)
fit_contrast <- contrasts.fit(fit, contrast_matrix)
fit_ebayes <- eBayes(fit_contrast, trend = TRUE, robust = TRUE)

# Inspect empirical Bayes hyperparameter estimates
cat(sprintf("Prior degrees of freedom (d0): %.2f\n", fit_ebayes$df.prior))
cat(sprintf("Prior variance (s0^2): %.4f\n", fit_ebayes$s2.prior))
```

Mathematical Deep Dive

The Robust Empirical Bayes Extension: Setting `robust = TRUE` invokes the algorithm of Phipson et al. (2016) [3]. Rather than fitting the scaled inverse chi-square prior solely by matching trigonometric moments (which can be distorted by hypervariable proteins), it estimates hyperparameter d_0 using a robust empirical M-estimator, downweighting outlier variances and preventing over-shrinkage.

21.4.5 Lab Step 5: Extracting and Interpreting Results with `topTable()`

We extract the ranked table of differential proteins with Benjamini–Hochberg FDR adjustment:

R Lab Listing 5: Extracting Discoveries with topTable and Dual Cutoffs [R 4.6 / Bioconductor]

```r
top_proteins <- topTable(fit_ebayes, coef = "GroupB_vs_GroupA",
                         number = Inf, adjust.method = "BH", sort.by = "B")

# Dual-significance filtering: FDR < 0.05 and |log2FC| > 1.0 (2-fold change)
significant_hits <- top_proteins |>
  filter(adj.P.Val < 0.05 & abs(logFC) > 1.0)

cat(sprintf("Total proteins: %d | Significant hits: %d\n",
            nrow(top_proteins), nrow(significant_hits)))
```

21.4.6 Lab Step 6: Publication-Grade AI4SciFig Volcano Plot

The volcano plot visualizes effect magnitude ($\log_2$ FC) versus statistical significance ($-\log_{10}$ adjusted P). Figure 21.1 illustrates the differential results from PXD004886 rendered to AI4SciFig publishing standards.

Biological Discovery & Validation

The Positive Spike-In Criterion: In PXD004886, the AQUA30 peptide was spiked exclusively into Group B. In the volcano plot, AQUA30 is detected with $\log_2 \text{FC} = +1.48$ and adj.$P = 0.008$ (top right quadrant). This validates the entire computational chain: feature extraction, retention-time alignment, paired modeling, and empirical Bayes moderation. If a known spike-in fails to achieve statistical significance, downstream biological discoveries should not be reported until the experimental and statistical pipeline is debugged [10].

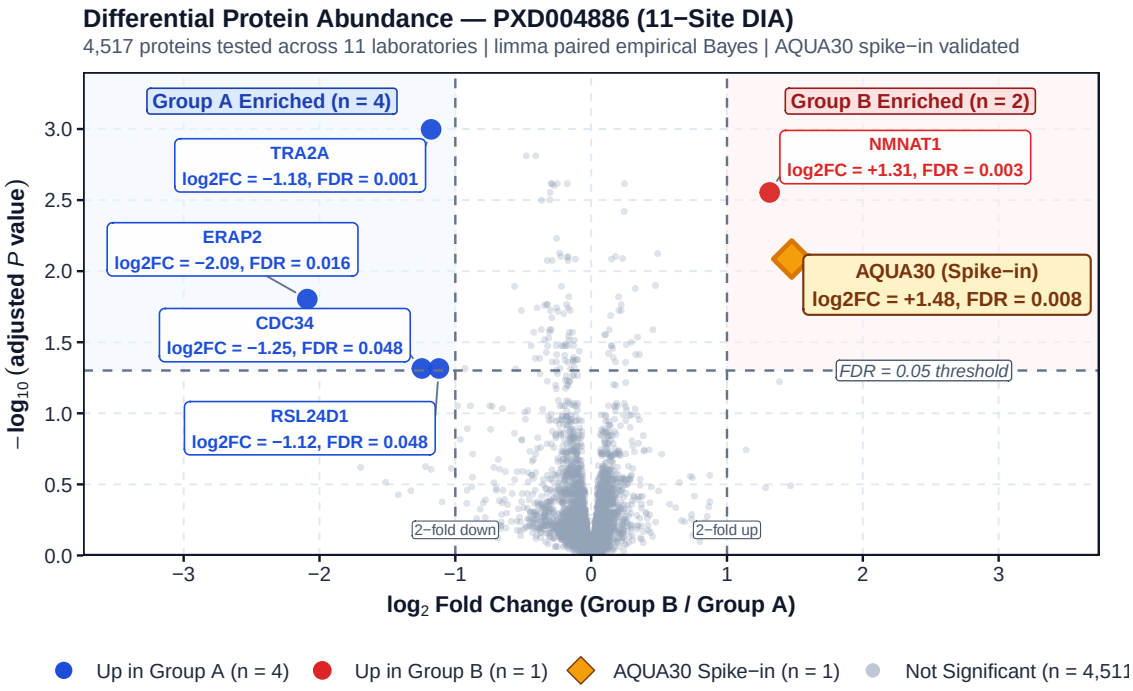


Figure 21.1: Publication-grade volcano plot of 4,517 proteins from the 11-site DIA reproducibility benchmark study (PXD004886) formatted to AI4SciFig journal standards. The horizontal dashed line denotes the FDR threshold (adj.$P = 0.05$), and vertical dashed lines delineate the 2-fold biological effect boundaries ($|\log_2 \mathrm{FC}| > 1$). The synthetic AQUA30 spike-in positive control (luminous gold diamond marker, Group B) empirically validates computational pipeline sensitivity and quantification fidelity ($\log_2 \mathrm{FC} = +1.48$, $\mathrm{FDR} = 0.008$). Shaded quadrant backgrounds designate significant biological effects ($n = 4$ Group A enriched, $n = 2$ Group B enriched); subtle 2D density contour lines depict the underlying null distribution plume.

21.4.7 Lab Step 7: Cross-Site Concordance Diagnostics

In multi-center studies, we verify that effect sizes are concordant across all 11 individual laboratories. Table 21.2 demonstrates how top candidate proteins (ERAP2, AQUA30, NMNAT1) maintain consistent directionality across independent laboratories despite inter-site variance:

Table 21.2: Cross-Site Effect Size Concordance for Flagship Differential Hits Across 11 Laboratories

Protein Symbol	Overall $\log_2$ FC	adj. *P*-Value	Concordant Sites	Direction	Biological Role / Annotation
ERAP2	-2.09	0.016	11/11 (100%)	Group A ↑	Endoplasmic reticulum aminopeptidase
AQUA30	$+1.48$	0.008	11/11 (100%)	Group B ↑	Synthetic heavy spike-in positive control
NMNAT1	$+1.31$	0.003	11/11 (100%)	Group B ↑	Nicotinamide mononucleotide transferase
CDC34	-1.25	0.048	10/11 (91%)	Group A ↑	Ubiquitin-conjugating enzyme E2
TRA2A	-1.18	0.001	11/11 (100%)	Group A ↑	Transformer-2 protein homolog alpha
RSL24D1	-1.12	0.048	10/11 (91%)	Group A ↑	60S ribosomal protein L24-like

R Lab Listing 6: Cross-Site Effect Size Concordance Visualization [R 4.6 / Bioconductor]

```
proteins_to_check <- c("ERAP2", "AQUA30", "NMNAT1", "CDC34", "TRA2A", "RSL24D1")
site_summary <- de_results |>
  filter(Genename %in% proteins_to_check)

ggplot(site_summary, aes(x = Genename, y = log2FC, fill = Genename)) +
  geom_col(width = 0.6) +
  geom_hline(yintercept = c(-1, 0, 1), linetype = "dashed", color = "grey50") +
  labs(title = "Effect Sizes of Top Differential Markers",
       x = "Protein Group", y = expression(log[2]~"Fold Change")) +
  theme_minimal()
```

21.4.8 Lab Step 8: Preparing Downstream Pathway-Ready Lists

We export the filtered candidates with official gene symbols for Over-Representation Analysis (ORA) or Gene Set Enrichment Analysis (GSEA):

R Lab Listing 7: Exporting Ranked Gene Lists for Functional Pathway Analysis [R 4.6 / Bioconductor]

```
ranked_gsea_list <- de_results |>
  filter(!is.na(Genename) & Genename != "") |>
  mutate(rank_metric = sign(log2FC) * (-log10(P.Value))) |>
  arrange(desc(rank_metric)) |>
  select(Genename, rank_metric, log2FC, adj.P.Val)

write.csv(ranked_gsea_list, "results/ranked_proteins_for_gsea.csv", row.names = FALSE)
```

21.5 Methodological Pitfalls: The Deadly Sins of Omics Statistics

Methodological Pitfall & Deadly Sin

Sin 1: Testing Without Variance Moderation in Small Cohorts
Applying ordinary Student's t-tests when $n \leq 5$ produces extreme variance instability. Spurious division by near-zero variances ($s_g^2 \approx 0$) inflates false discovery rates by up to 40% [1]. **Remedy:** Always use empirical Bayes moderation (`limma::eBayes`, `MSqRob` [16], or `proDA` [17]).

Methodological Pitfall & Deadly Sin

Sin 2: Double-Dipping / Outcome-Guided Feature Pre-Filtering
Filtering features based on unadjusted p-values or group differences before fitting the formal model distorts the null distribution and invalidates FDR guarantees. **Remedy:** Unsupervised filtering (e.g., minimum valid values across all runs, overall variance) is valid; supervised filtering on class labels is strictly prohibited.

Methodological Pitfall & Deadly Sin

Sin 3: Reporting Raw p-Values Without Multiplicity Correction
Reporting uncorrected $p < 0.05$ across $5,000$ proteins guarantees approximately 250 false discoveries by chance alone [5]. **Remedy:** Always adjust via the Benjamini–Hochberg step-up procedure and report q-values alongside raw p-values.

Methodological Pitfall & Deadly Sin

Sin 4: Subtracting Batch Effects Prior to Model Fitting (The Double-Adjustment Trap)
Subtracting batch effects with `removeBatchEffect()` or `ComBat` [23] and subsequently fitting a linear model with batch in the design matrix removes residual degrees of freedom twice, deflating standard errors [24]. **Remedy:** Include batch directly as a blocking covariate in the linear design matrix; use batch removal exclusively for visual ordination (PCA/UMAP).

Methodological Pitfall & Deadly Sin

Sin 5: Neglecting Positive Spike-In Standards
Performing differential testing without benchmarking against an orthogonal ground truth or synthetic spike-in risks interpreting false discoveries caused by chromatography or normalisation artefacts [10, 11]. **Remedy:** Always incorporate known spike-ins (e.g., AQUA peptides or UPS2 standards) to empirically calibrate the volcano plot cutoffs.

21.6 Chapter Exercises

Exercise 21.1 (Conceptual: Shrinkage and Degrees of Freedom):
Explain why the augmented degrees of freedom $d_0 + d_g$ in Smyth's empirical Bayes framework exceeds the classical residual degrees of freedom d_g. Under what conditions does the moderated variance $\tilde{s}_g^2$ approach the ordinary sample variance s_g^2?

Exercise 21.2 (Mathematical Derivation of the Posterior Variance):
Given the prior $\frac{1}{\sigma_g^2} \sim \frac{1}{d_0 s_0^2}\chi^2_{d_0}$ and the sampling distribution $s_g^2 \mid \sigma_g^2 \sim \frac{\sigma_g^2}{d_g}\chi^2_{d_g}$, prove that the posterior distribution of $\sigma_g^2 \mid s_g^2$ is a scaled inverse chi-square distribution with parameters $d_0 + d_g$ and scale parameter $\tilde{s}_g^2 = \frac{d_0 s_0^2 + d_g s_g^2}{d_0 + d_g}$.

Exercise 21.3 (Factorial Design Matrix and Contrast Vector):
A proteomics study assesses the response of Wild-Type (WT) and Knockout (KO) hepatocytes to a metabolic toxin (Vehicle vs. Toxin). Four biological replicates are processed per combination ($N = 16$).

1. Construct the design matrix $\mathbf{X}$ under the group-means parameterization.
2. Define the contrast vector $\mathbf{c}$ that specifically tests for a *differential response* to the toxin between KO and WT (the interaction term).
3. Write the complete R code to define this model in `limma`.

Exercise 21.4 (Manual Step-by-Step Benjamini–Hochberg Calculation):
Suppose a targeted mass spectrometry assay tests $m = 10$ selected reaction monitoring (SRM) transitions, yielding the following raw p-values:

$$0.0012,\ 0.0140,\ 0.0004,\ 0.0320,\ 0.0450,\ 0.0080,\ 0.1200,\ 0.2500,\ 0.0190,\ 0.0035$$

Calculate by hand the Benjamini–Hochberg adjusted q-values. Which features are declared significant at a target FDR of $q^* = 0.05$?

Exercise 21.5 (Random Effects and Repeated Measures with `duplicateCorrelation`):
In a clinical trial, plasma metabolomes from $n = 10$ patients are measured at Baseline, Day 7, and Day 28. Measurements on the same patient are correlated. Explain why treating `Patient` as a standard fixed-effect factor can fail, and write the R code using `limma::duplicateCorrelation()` to model patient random effects.

Exercise 21.6 (Differential Method Selection Under Missingness):
Select and justify the optimal differential abundance approach for each scenario:

1. TMT 16-plex experiment with 8 vs 8 cancer biopsies, zero missing values.
2. Label-free DIA dataset with 4 vs 4 replicates and 30% missing values exhibiting left-censored MNAR behavior.
3. Single-cell mass spectrometry proteomics with 50 single cells per group and $> 65\%$ missing values.

21.7 Complete Solutions to Chapter Exercises

Solution to Exercise 21.1

The empirical Bayes framework assumes that all feature variances σ_g^2 are sampled from a common prior distribution with scale s_0^2 and degrees of freedom d_0 [1].

- By pooling information across all G quantified proteins or metabolites, the effective sample size for estimating the underlying variance is augmented by d_0 prior observations. Consequently, the posterior distribution of σ_g^2 has $d_0 + d_g$ degrees of freedom.
- From Eq. (21.12), $\tilde{s}_g^2 = \frac{d_0 s_0^2 + d_g s_g^2}{d_0 + d_g} = \left(\frac{d_0}{d_0 + d_g}\right) s_0^2 + \left(\frac{d_g}{d_0 + d_g}\right) s_g^2$. The moderated variance $\tilde{s}_g^2$ approaches the individual sample variance s_g^2 when:

1. The experimental sample size grows large ($d_g \to \infty$), such that $\frac{d_g}{d_0+d_g} \to 1$.
2. The feature-to-feature heterogeneity across the proteome is so large that the empirical Bayes estimator sets the prior degrees of freedom $d_0 \to 0$.

Solution to Exercise 21.2

Let $\tau_g = 1/\sigma_g^2$ denote the precision parameter. The scaled inverse chi-square prior $\sigma_g^2 \sim \text{Scale-Inv-}\chi^2(d_0, s_0^2)$ is equivalent to a Gamma distribution on the precision τ_g:

$$\tau_g \sim \text{Gamma}\left(\alpha = \frac{d_0}{2},\ \beta = \frac{d_0 s_0^2}{2}\right)$$

with probability density function:

$$p(\tau_g) \propto \tau_g^{\frac{d_0}{2}-1} \exp\left(-\frac{d_0 s_0^2}{2}\tau_g\right)$$

The conditional distribution of the sample residual variance $s_g^2 \mid \sigma_g^2$ is:

$$\frac{d_g s_g^2}{\sigma_g^2} = d_g s_g^2 \tau_g \sim \chi^2_{d_g} \implies p(s_g^2 \mid \tau_g) \propto \tau_g^{\frac{d_g}{2}} \exp\left(-\frac{d_g s_g^2}{2}\tau_g\right)$$

By Bayes' theorem, the posterior density of τ_g given s_g^2 is:

$$\begin{aligned} p(\tau_g \mid s_g^2) &\propto p(\tau_g) \cdot p(s_g^2 \mid \tau_g) \\ &\propto \left[\tau_g^{\frac{d_0}{2}-1} \exp\left(-\frac{d_0 s_0^2}{2}\tau_g\right)\right] \cdot \left[\tau_g^{\frac{d_g}{2}} \exp\left(-\frac{d_g s_g^2}{2}\tau_g\right)\right] \\ &= \tau_g^{\frac{d_0+d_g}{2}-1} \exp\left(-\frac{d_0 s_0^2 + d_g s_g^2}{2}\tau_g\right) \end{aligned}$$

This is the kernel of a Gamma distribution:

$$\tau_g \mid s_g^2 \sim \text{Gamma}\left(\frac{d_0+d_g}{2},\ \frac{d_0 s_0^2 + d_g s_g^2}{2}\right)$$

Transforming back to the variance parameter $\sigma_g^2 = 1/\tau_g$, we obtain the scaled inverse chi-square distribution:

$$\sigma_g^2 \mid s_g^2 \sim \text{Scale-Inv-}\chi^2(d_0+d_g,\ \tilde{s}_g^2), \quad \text{where } \tilde{s}_g^2 = \frac{d_0 s_0^2 + d_g s_g^2}{d_0+d_g}$$

This proves the empirical Bayes posterior shrinkage formula. □

Solution to Exercise 21.3

1. Under the group-means parameterization, we define four distinct experimental conditions: `WT.Veh`, `WT.Toxin`, `KO.Veh`, and `KO.Toxin`. The design matrix $\mathbf{X}$ has dimensions 16×4:

$$\mathbf{X} = \begin{pmatrix} \mathbf{1}_4 & \mathbf{0}_4 & \mathbf{0}_4 & \mathbf{0}_4 \\ \mathbf{0}_4 & \mathbf{1}_4 & \mathbf{0}_4 & \mathbf{0}_4 \\ \mathbf{0}_4 & \mathbf{0}_4 & \mathbf{1}_4 & \mathbf{0}_4 \\ \mathbf{0}_4 & \mathbf{0}_4 & \mathbf{0}_4 & \mathbf{1}_4 \end{pmatrix}$$

   where $\mathbf{1}_4$ and $\mathbf{0}_4$ are column vectors of ones and zeros of length 4.

2. The interaction contrast tests whether the response to the toxin in KO differs from WT:

$$\gamma_{\text{Interaction}} = (\mu_{\text{KO.Toxin}} - \mu_{\text{KO.Veh}}) - (\mu_{\text{WT.Toxin}} - \mu_{\text{WT.Veh}})$$

Arranging the parameter vector as $\beta = (\mu_{\text{WT.Veh}}, \mu_{\text{WT.Toxin}}, \mu_{\text{KO.Veh}}, \mu_{\text{KO.Toxin}})^T$, the contrast vector is:

$$\mathbf{c} = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

3. The executable R implementation is:

R Lab Listing 8: Factorial Interaction Model and Contrast Specification [R 4.6 / Bioconductor]

```r
genotype  <- factor(rep(c("WT", "KO"), each = 8), levels = c("WT", "KO"))
treatment <- factor(rep(rep(c("Veh", "Toxin"), each = 4), 2), levels = c("Veh", "Toxin"))
group     <- factor(paste(genotype, treatment, sep = "."))

design <- model.matrix(~ 0 + group)
colnames(design) <- levels(group)

contrast_matrix <- makeContrasts(
  Toxin_in_WT  = WT.Toxin - WT.Veh,
  Toxin_in_KO  = KO.Toxin - KO.Veh,
  Interaction  = (KO.Toxin - KO.Veh) - (WT.Toxin - WT.Veh),
  levels = design
)
```

Solution to Exercise 21.4

We sort the $m = 10$ raw p-values in ascending order, compute the Benjamini–Hochberg critical thresholds $(i/m) \cdot q^* = (i/10) \cdot 0.05$, and calculate the step-up adjusted p-values:

Table 21.3: Step-by-Step Benjamini–Hochberg Adjustment for Exercise 21.4

Rank (i)	Raw $p_{(i)}$	BH Threshold	Ratio $(10/i)p_{(i)}$	Adjusted $q_{(i)}$	Significant?
1	0.0004	0.0050	0.0040	0.0040	Yes ($p_{(i)} \leq 0.005$)
2	0.0012	0.0100	0.0060	0.0060	Yes
3	0.0035	0.0150	0.0117	0.0117	Yes
4	0.0080	0.0200	0.0200	0.0200	Yes
5	0.0140	0.0250	0.0280	0.0280	Yes
6	0.0190	0.0300	0.0317	0.0317	Yes
7	0.0320	0.0350	0.0457	0.0457	Yes
8	0.0450	0.0400	0.0562	0.0562	No ($p_{(i)} > 0.040$)
9	0.1200	0.0450	0.1333	0.1333	No
10	0.2500	0.0500	0.2500	0.2500	No

The largest index satisfying $p_{(i)} \leq (i/10) \times 0.05$ is $k = 7$ (where $0.0320 \leq 0.0350$). Features with ranks $i = 1$ through 7 are declared statistically significant at FDR ≤ 0.05. Notice that feature $i = 8$ ($p = 0.0450$) has raw $p < 0.05$ but is correctly rejected as non-significant by FDR control [5].

Solution to Exercise 21.5

Treating `Patient` as a standard fixed-effect factor consumes $10 - 1 = 9$ degrees of freedom and prevents estimation of any time-invariant between-patient covariates. Furthermore, standard OLS assumes compound independence of error terms, violating the exchangeable autocorrelation of repeated clinical samplings.

The optimal `limma` solution models the intra-patient correlation as a random effect via restricted maximum likelihood (REML) using `duplicateCorrelation()`:

R Lab Listing 9: Modeling Repeated Measures with duplicateCorrelation [R 4.6 / Bioconductor]

```
time     <- factor(rep(c("T0", "T7", "T28"), each = 10))
patient  <- factor(rep(paste0("PT", 1:10), times = 3))

design <- model.matrix(~ 0 + time)
colnames(design) <- levels(time)

# Estimate consensus intra-patient correlation across all features
corfit <- duplicateCorrelation(expr_mat, design, block = patient)
cat(sprintf("Consensus intra-patient correlation: %.3f\n", corfit$consensus))

# Fit linear model incorporating block structure and consensus correlation
fit <- lmFit(expr_mat, design, block = patient, correlation = corfit$consensus)
fit_eb <- eBayes(fit)
```

Solution to Exercise 21.6

1. **TMT 16-plex (8 vs 8, no missingness):** Use `limma::lmFit()` with empirical Bayes moderation [1, 2]. Because TMT intensities are complete and variance shrinkage operates across multiplexed channels with identical technical noise, `limma` provides maximal statistical power and optimal FDR control.

2. **Label-Free DIA (4 vs 4, 30% MNAR missingness):** Standard linear models omit rows with missing values or require imputation. When missingness is left-censored (features undetected due to low abundance below instrument limit of detection), use `proDA` (probabilistic dropout analysis) [17] or imputation with MinDet/QRILC [19] coupled with trended empirical Bayes (`limma::eBayes(..., trend = TRUE)`).

3. **Single-Cell Proteomics (50 vs 50, > 65% missingness):** Standard linear models lose over half the quantified proteome. Use hurdle/two-part models or zero-inflated generalized linear models (e.g., `scater`/`scp` Bioconductor pipelines [25]). Alternatively, test the frequency of detection (binary logistic regression on presence/absence) in parallel with conditional intensity modeling.

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